[A]**2.3** Answer each part for the following context-free grammar G.

$$
\begin{aligned}
R &\to XRX \mid S \\
S &\to \mathtt{a}T\mathtt{b} \mid \mathtt{b}T\mathtt{a} \\
T &\to XTX \mid X \mid \varepsilon \\
X &\to \mathtt{a} \mid \mathtt{b}
\end{aligned}
$$

a. What are the variables of G?
b. What are the terminals of G?
c. Which is the start variable of G?
d. Give three strings in $L(G)$.
e. Give three strings *not* in $L(G)$.
f. True or False: $T \Rightarrow \mathtt{aba}$.
g. True or False: $T \overset{*}{\Rightarrow} \mathtt{aba}$.
h. True or False: $T \Rightarrow T$.
i. True or False: $T \overset{*}{\Rightarrow} T$.
j. True or False: $XXX \overset{*}{\Rightarrow} \mathtt{aba}$.
k. True or False: $X \overset{*}{\Rightarrow} \mathtt{aba}$.
l. True or False: $T \overset{*}{\Rightarrow} XX$.
m. True or False: $T \overset{*}{\Rightarrow} XXX$.
n. True or False: $S \overset{*}{\Rightarrow} \varepsilon$.
o. Give a description in English of $L(G)$.

2.4 Give context-free grammars that generate the following languages. In all parts, the alphabet Σ is $\{0,1\}$.

[A]**a.** $\{w \mid w$ contains at least three 1s$\}$
b. $\{w \mid w$ starts and ends with the same symbol$\}$
c. $\{w \mid$ the length of w is odd$\}$
[A]**d.** $\{w \mid$ the length of w is odd and its middle symbol is a 0$\}$
e. $\{w \mid w = w^{\mathcal{R}}$, that is, w is a palindrome$\}$
f. The empty set

2.5 Give informal descriptions and state diagrams of pushdown automata for the languages in Exercise 2.4.

2.6 Give context-free grammars generating the following languages.

[A]**a.** The set of strings over the alphabet $\{\mathtt{a},\mathtt{b}\}$ with more a's than b's
b. The complement of the language $\{\mathtt{a}^n\mathtt{b}^n \mid n \geq 0\}$
[A]**c.** $\{w\#x \mid w^{\mathcal{R}}$ is a substring of x for $w, x \in \{0,1\}^*\}$
d. $\{x_1\#x_2\#\cdots\#x_k \mid k \geq 1$, each $x_i \in \{\mathtt{a},\mathtt{b}\}^*$, and for some i and j, $x_i = x_j^{\mathcal{R}}\}$

[A]**2.7** Give informal English descriptions of PDAs for the languages in Exercise 2.6.

[A]**2.8** Show that the string `the girl touches the boy with the flower` has two different leftmost derivations in grammar G_2 on page 103. Describe in English the two different meanings of this sentence.

2.9 Give a context-free grammar that generates the language

$$A = \{\mathtt{a}^i\mathtt{b}^j\mathtt{c}^k \mid i = j \text{ or } j = k \text{ where } i, j, k \geq 0\}.$$

Is your grammar ambiguous? Why or why not?

2.10 Give an informal description of a pushdown automaton that recognizes the language A in Exercise 2.9.

2.11 Convert the CFG G_4 given in Exercise 2.1 to an equivalent PDA, using the procedure given in Theorem 2.20.

2.12 Convert the CFG G given in Exercise 2.3 to an equivalent PDA, using the procedure given in Theorem 2.20.

2.13 Let $G = (V, \Sigma, R, S)$ be the following grammar. $V = \{S, T, U\}$; $\Sigma = \{0, \#\}$; and R is the set of rules:

$$S \to TT \mid U$$
$$T \to 0T \mid T0 \mid \#$$
$$U \to 0U00 \mid \#$$

a. Describe $L(G)$ in English.

b. Prove that $L(G)$ is not regular.

2.14 Convert the following CFG into an equivalent CFG in Chomsky normal form, using the procedure given in Theorem 2.9.

$$A \to BAB \mid B \mid \varepsilon$$
$$B \to 00 \mid \varepsilon$$

2.15 Give a counterexample to show that the following construction fails to prove that the class of context-free languages is closed under star. Let A be a CFL that is generated by the CFG $G = (V, \Sigma, R, S)$. Add the new rule $S \to SS$ and call the resulting grammar G'. This grammar is supposed to generate A^*.

2.16 Show that the class of context-free languages is closed under the regular operations, union, concatenation, and star.

2.17 Use the results of Exercise 2.16 to give another proof that every regular language is context free, by showing how to convert a regular expression directly to an equivalent context-free grammar.

PROBLEMS

2.18 Consider the following CFG G:

$$S \to SS \mid T$$
$$T \to \mathtt{a}T\mathtt{b} \mid \mathtt{ab}$$

Describe $L(G)$ and show that G is ambiguous. Give an unambiguous grammar H where $L(H) = L(G)$ and sketch a proof that H is unambiguous.

***2.19** We defined the rotational closure of language A to be $RC(A) = \{yx \mid xy \in A\}$. Show that the class of CFLs is closed under rotational closure.

***2.20** We defined the *CUT* of language A to be $CUT(A) = \{yxz \mid xyz \in A\}$. Show that the class of CFLs is not closed under *CUT*.

2.21 Show that every DCFG is an unambiguous CFG.

A***2.22** Show that every DCFG generates a prefix-free language.

*2.31 Let CFG G be the following grammar.

$$
\begin{aligned}
S &\rightarrow \mathtt{a}S\mathtt{b} \mid \mathtt{b}Y \mid Y\mathtt{a} \\
Y &\rightarrow \mathtt{b}Y \mid \mathtt{a}Y \mid \varepsilon
\end{aligned}
$$

Give a simple description of $L(G)$ in English. Use that description to give a CFG for $\overline{L(G)}$, the complement of $L(G)$.

2.32 Let $A/B = \{w \mid wx \in A \text{ for some } x \in B\}$. Show that if A is context free and B is regular, then A/B is context free.